BIG IDEAS MATH • GRADE 7 • CHAPTER 8

# Circles & Area: Circumference, Pi & Modulo Dial

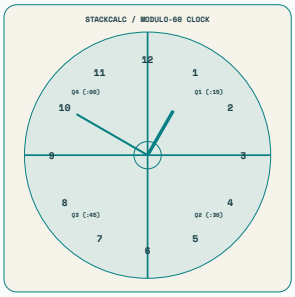
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.7.G.B.4

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

**STUDENT SELF-CHECK RULE (NO CHEATING):**
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

## STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

### 1. 2D Bench Blueprint




Kit: Clock Dial

### 2. Desk Setup (Concrete -> Pictorial)

**Physical Bench Setup:**
Wrap string around circular dial; stretch across 3+ diameters to discover ratio π ≈ 3.1416.

**Sketch Model on Paper:**
Draw circle showing radius r, diameter d = 2r, and area sectors unrolled into parallelogram.

### 3. Step-by-Step LCD Stack Trace (SC-32)

MODE: 123 Base-10 Arithmetic

| | | | |
|---|---|---|---|
| 1. | 5 | X: 5 | Radius r = 5 into X |
| 2. | x² | X: 25 | Square radius: 5² = |
| 3. | π | Y: 25 \| X: 3.1416 | Constant π into X |
| 4. | × | X: 78.5398 | Area: 25 × π = 78.53 |

EXPECTED LCD X: Area = 78.5398 cm²

**1** **Find circumference of circle with diameter d = 10 cm: 10 × π.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-32

**2** **Find area of circle with radius r = 4 cm: 4² × π.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-32

**3** **Find area of circle with radius r = 7 cm: 7² × π.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-32

**4** **Find radius of circle with circumference C = 62.83 cm.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-32

# Station Diagnostic & Solution Key: Chapter 8

Circles & Area: Circumference, Pi & Modulo Dial • Manipulative Setup, Diagnostic Rubric & Verification Flow

## MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

**Station Kit: 4-Quadrant Modulo Clock & Rotary Dial**

CAD Blueprint: clock_dial

**Target Device: SC-32 Scientific**

**CAD / STL Source File:**

Education/scad/lib/clock_quadrants.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

**Recommended Filaments:**

White Faceplate with Oceanic Teal Hands (PLA)

| # | Math Problem Prompt | Expected Math Answer | Self-Check Keystrokes & RPN Stack |
|---|---|---|---|
| 1 | **Find circumference of circle with diameter d**<br>Standard: CCSS.MATH.7.G.B.4 | **31.4159 cm** | [1][0] [ENTER] [π] [×] |
| 2 | **Find area of circle with radius r = 4 cm: 4²**<br>Standard: CCSS.MATH.7.G.B.4 | **50.2655 cm²** | [4] [x²] [π] [×] |
| 3 | **Find area of circle with radius r = 7 cm: 7²**<br>Standard: CCSS.MATH.7.G.B.4 | **153.9380 cm²** | [7] [x²] [π] [×] |
| 4 | **Find radius of circle with circumference C =**<br>Standard: CCSS.MATH.7.G.B.4 | **r ≈ 10.0 cm** | [6][2][.][8][3] [ENTER] [2] [÷] [π] [÷ |

## TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

**Key Misconception to Target During Station Observation:**

Watch for confusing diameter with radius when calculating area πr².

**How the Manipulatives & Calculator Foster Self-Correction:**

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

**Station Facilitation & Pacing Tip:**

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

## CLASSROOM SCORING & STATION MASTERY RUBRIC

| | |
|---|---|
| **Exemplary (18–20 pts):** | Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack. |
| **Proficient (14–17 pts):** | Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting. |
| **Developing (10–13 pts):** | Understands problem concept but made computational slips; successfully self-corrected using calculator LCD. |
| **Intervention (<10 pts):** | Relied on calculator before solving on paper, or struggled with underlying physical manipulative models. |